

IMS568 – INFORMATION MANAGEMENT SYSTEM PROJECT (CAPSTONE)

OUTLINE SAMPLE OF CAPSTONE PROJECT PROPOSAL REPORT

1. Introduction

Provide an overview of the project, stating the motivation behind choosing the system to be developed. Highlight the system type (desktop / web-based / data visualization application) and the primary problem it aims to solve or improve. Mention the scope of the project and its relevance to real-world applications.

2. Problem Description

Clearly define the problem your system seeks to address. Discuss the limitations or inefficiencies in the current situation or existing solutions, and explain why there is a need for a new system. Include specific examples or case studies to substantiate the problem.

3. Objectives

List the specific goals and objectives of your system. These should be clear and measurable. Examples include improving efficiency, automating a process, enhancing user experience, or creating a system with specific features. You need to list the objectives (2-3) that you wish to achieve at the end of the capstone project. At the end of the final evaluation of the capstone project, these objectives will be used to judge your project completion status.

Example:

- To develop a user-friendly interface for booking appointments.
 - To design a secure database for managing user data efficiently.
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4. Literature Survey & Benchmarking

Conduct a review of existing literature, similar systems, or technologies that have been developed to address similar problems. Compare various approaches and identify the strengths and weaknesses of these systems. This section should provide insight into existing solutions and justify the need for your system development.

Benchmarking:

As part of your literature review, conduct a benchmarking comparison between your proposed system and existing systems (at least three systems). Analyze how your system measures up in terms of features, performance, security, usability, or efficiency. Highlight the key indicators and performance metrics that will be used to evaluate your system against these existing solutions.

Example comparison points:

- Additional features offered by your system that are not offered by other benchmarked systems
 - How your system improves upon speed or user experience compared to current alternatives.
 - Performance metrics such as load times, data processing capacity, or security protocols.
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5. Methodology

This section explains the approach you will use to develop the system, including the methodology you have chosen (e.g., Waterfall, Agile, Scrum, etc.). Justify why this methodology is the most suitable for your project.

Key components to include:

- **System Design:** Outline how the system will be designed, including interface layouts, data flow diagrams, and system architecture diagrams.
 - **Database Design:** Include **Entity Relationship Diagrams (ERD)** and **Database Schema** to show the structure of the database.
 - **Data Dictionary:** Provide a detailed data dictionary listing all the tables, fields, data types, and their descriptions.
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6. Work Plan

Present a detailed timeline or **Gantt chart** showing the phases of the project from initiation to completion. Include milestones such as research, design, development, testing, and deployment. Ensure that the plan is feasible and covers all critical stages of system development.

7. Conclusion

Summarize the importance of the project and its potential contributions. Restate how your system will address the identified problem, how it stands out from other solutions, and its anticipated impact.

8. References

Provide a complete list of all references and sources used in your research, including books, journal articles, websites, and other documents cited in the literature survey. Follow a consistent citation style (APA, MLA, etc.).
